Double angle identities

1

a $\frac{\sqrt{3}}{2}$ b $\frac{\sqrt{2}}{2}$ c $\frac{\sqrt{2}}{2}$ d $\frac{\sqrt{3}}{2}$

e $\frac{5\sqrt{2}}{2}$ f $-\frac{\sqrt{2}}{24}$ g $\frac{3\sqrt{3}}{2}$ h $-\frac{\sqrt{2}}{8}$

2 $-\frac{5\sqrt{2}}{2}$

3

a $\sin\theta = -\frac{3}{5}$ b $-\frac{24}{25}$ c $\frac{7}{25}$

4

a $\cos\theta = -\frac{24}{25}$ b $\frac{336}{625}$ c $\frac{527}{625}$

5

a $\frac{24}{25}$ b $\frac{7}{25}$

6

a $\cos^2\left(\frac{x}{2}\right)$ b $\cos x = \frac{5}{13}$ c $-\frac{3\sqrt{13}}{13}$

7

a $\cos^2\left(\frac{x}{2}\right)$ b $-\frac{12}{13}$ c $\frac{\sqrt{26}}{26}$

8

a $\sin^2\left(\frac{x}{2}\right)$ b $\frac{\sqrt{17}}{17}$ c $\frac{\sqrt{578-34\sqrt{17}}}{34}$

9

a $\frac{\sqrt{2+\sqrt{2}}}{2}$ b $-\frac{\sqrt{2-\sqrt{3}}}{2}$ c $\frac{\sqrt{2+\sqrt{2}}}{2}$ d $-\frac{\sqrt{2+\sqrt{2}}}{2}$

10

a $\frac{\sqrt{154}}{14}$ b $\cos\left(\frac{x}{2}\right) = \frac{\sqrt{70}}{10}$ c $-\frac{\sqrt{30}}{6}$ d $-\frac{\sqrt{154}}{14}$

11 q

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12 **a** No solution provided.

b $\sin 18^\circ = \dfrac{4\cos^2(18^\circ) - 3}{2}$

13

a No solution provided.

b No solution provided.

c No solution provided.

d No solution provided.

e No solution provided.

f No solution provided.

Equations

14

a $x = \dfrac{\pi}{3}, \dfrac{5\pi}{3}, 0$

b $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}, \dfrac{\pi}{6}, \dfrac{5\pi}{6}$

c $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}, \dfrac{\pi}{2}$

d $x = 128.2^\circ, 231.8^\circ$

e $x = 90^\circ, 270^\circ, 135^\circ, 315^\circ$

15

a No solution provided.

b $\theta = 0, \pi, \dfrac{\pi}{6}, \dfrac{5\pi}{6}$

16

a $1 - \sin 2\theta$

b The equation has at least 1 real solution.

Applications

17

a $\theta = 84$

b $m = 1.4$

18

a $\dfrac{755}{4}\sin^2(110\pi t)$

b $\text{Maximum} = \dfrac{755}{4}$

$\text{Minimum} = 0$

c $-\dfrac{755}{8}\cos 220\pi t + \dfrac{755}{8}$

d 90 W

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19

a $x = \cos a$

b $y = \sin a$

c No solution provided

d $a = 45^\circ$

e $\frac{\sqrt{2}}{2}$

f $\sqrt{2}$

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